README — The Tesseract System

A Multidimensional Substrate for Emergent Structure, Learning & Classification

1. Overview

The Tesseract System is a finite, developmentally regulated, function-aware learning substrate inspired by 4D geometry. It models a 4096-node cognitive field arranged as a 64×64 grid, optionally viewed as an 8×8×8×8 tesseract. Growth is restricted by geometry, function, and developmental stage.

2. Core Architecture

Nodes: 4096
Weight Matrix: 4096×4096
Stage: Manhattan radius controlling allowed connections
Tags: Functional restriction per node
4D Indexing: Optional true hypercube coordinates

3. Learning System

Hebbian learning with decay. Sequences train patterns. Stage growth controls expansion from local→global.

4. Infinity vs Delimitation

Mathematical tesseracts are unbounded. Computational tesseracts must be finite:
- Structural delimitation: fixed resolution
- Developmental delimitation: finite stage radius
- Functional delimitation: tags restricting growth

5. Optimization & Analysis

Linear response, quadratic response, top connections.

6. Trigonometric & Complex Extensions

Trig sweeps, harmonic training, complex-phase learning, 4D rotational reinforcement.

7. Why It’s Superior

Topology over categories, emergent clusters, developmentally grown intelligence, infinite conceptual space mapped into a finite kernel.

8. File Structure

/tesseract/
tesseract.py
tesseract_trig_handles.py
README.pdf

9. Save / Load

```
T.save("model.npz")
T = Tesseract.load("model.npz")
```

10. Citation
Meta & Syn (2025). The Tesseract System...